

COMP6900 Major Project Scoping Report Section

Smart

Produce

Scanner

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1. Project Statement

1.1 Overview of the Idea

The Smart Produce Scanner is a web browser-based application that will be able to identify some fruit from an image and give information regarding the identified fruit. Initial scope is apples and bananas. A user uploads a picture using the web application, the picture pre-processed for the computational process, and a trained Convolutional Neural Network (CNN) is used to classify it as apple or banana. Once classified, the purpose of the application is to offer practical information including nutritional information, health information, ripeness and possible uses.

The proposed workflow is simple, which includes the uploading of images, pre-processing of images, classification of CNN, and fruit identification. CNNs can be used for this purpose since they can be trained directly from images, capturing visual features. Convolutional layers are used to identify features like edges, shapes, colors and textures, and pooling is used to decrease feature maps and computation. These learned features are then fused by fully connected layers and the output layer generates probabilities of the classes available (O'Shea & Nash, 2015).

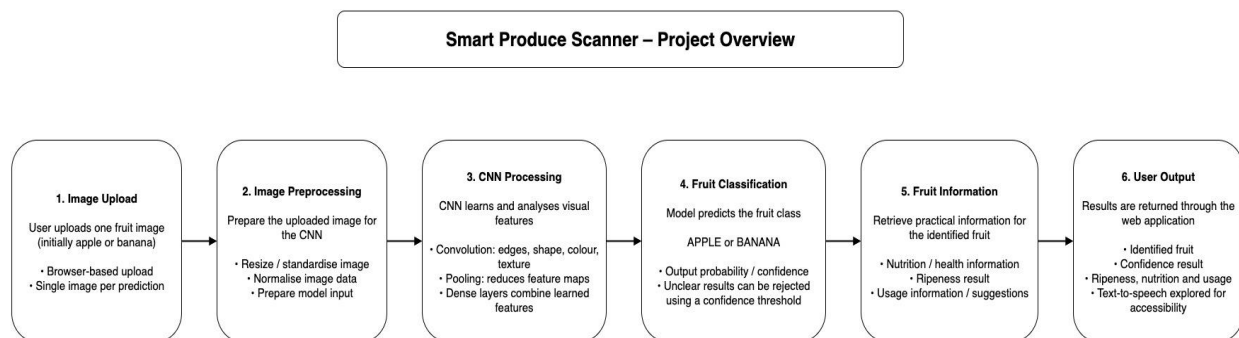


Figure 1: Smart Produce Scanner- Workflow

The project is deliberately a small prototype. It doesn't have a mobile app and only analyzes one fruit from a single photo at this stage, not multiple fruits or live video. This provides the team with a working platform which can be expanded to other fruits and characteristics.

2. The Project

2.1 Needs and Problems

The project answers this question by implementing an image recognition system that can be used to recognize some fruit and then return useful information to the users by using one application. In addition to fruit identification, users may wish to get nutrition information, ripeness, health benefits, and uses information about the fruit. If the application is not integrated, users might have to determine the fruit first and then look elsewhere for further data.

Another difficulty is that real world fruit photos aren't taken under the same conditions. The lighting, camera angle, background and appearance of the fruit can be quite different. Therefore, a variety of apple and banana images should be used for training the model. The team will capture and archive images in various contexts and add appropriate public image datasets. This should allow the model to be able to learn general visual attributes instead of a small number of image conditions.

The project is therefore not only to write another fruit classifier but to design a small usable prototype of an image recogniser which links image recognition to useful fruit information. The project also offers the team a chance to implement CNN concepts with a real application and assess the coordination between the training of the models, the preparation of the data, the integration with a web and the user output.

2.2 Has This Been Addressed Before?

- **Established field:** The CNN-based fruit and produce recognition software has already been established, with public datasets and numerous commercial apps like "can your produce" (Mureşan & Oltean 2018) and "O'Shea & Nash (2015)".
- **Gap:** Despite being the most widely used tools for species identification, few incorporate data-driven evaluation methods such as spoken nutrition and usage output into one pipeline.
- **Accessibility gap:** Scan-and-identify apps allow users to see and read a screen, which is essential for blind individuals who relied on low vision experiences.
- **Proposed innovation:** The system is a process that categorizes fruits, grades them ripe and provides natural speech. Blind viewers are given a complete screen-independent conversational output rather than an on-screen prediction, as it operates from the end to achieve this goal.
- **Where the value sits:** In integrating and usability, the best approach is to use an actual representation of a real-world case or scenario that can be tested on various devices. This proves that not every single component in our systems is

2.3 Target Audience

- **Primary:** Blind or low-sighted individuals who rely on spoken output for fruit identification, ripeness, nutrition and usage are more proficient at using written output than reading text from the screen.
- **Secondary:** Users who wish to observe a fruit's freshness, nutritional content or utilization are encouraged to use their sight frequently.
- **Tertiary:** Consuming a balanced diet or managing specific conditions with caution is an effective way of managing health issues. Diabetes and students or hobbyists interested in practical CNN applications can be summarized as diabetes, and hobbyist, who are interested in the use of
- **Out of scope:** commercial or industrial sorting use cases (e.g. supermarket QA) this is a consumer-facing tool, not an industrial classifier.

- **Context of use:** casual, everyday settings such as home kitchens and grocery shopping, accessed via a standard smartphone or laptop browser rather than specialised hardware.

2.4 Ethical Impacts and Considerations

- **Data privacy:** uploaded images are processed only for classification; no images are stored or linked to identity beyond the session.
- **Consent & dataset provenance:** If individuals or entities appear in self-collected images, they must obtain permission and should be checked for licensing terms.
- **Bias & fairness:** the training set must represent varied lighting, backgrounds and fruit varieties so the model doesn't underperform on less-represented conditions.
- **Misclassification risk:** a wrong classification could give misleading dietary information, mitigated by the confidence threshold that rejects unclear images.
- **Transparency:** users are told the system is a prototype, probabilistic classifier, not a certified food-safety or nutrition tool.

3. Project Goal and Objectives

3.1 Goal

Build an image identification model that can identify fruits from a given file, you can start by selecting the apple and banana as primary variables and then add more variety to the list. The system determines a particular outcome and delivers the result, along with nutritional value and usage information, as spoken output.

3.2 Objectives

- Collect and label 2,000+ images across varied lighting conditions and backgrounds, to give the classifier a representative training set.
- Develop a Convolutional Neural Network (CNN) which is effective in not only accurately identifying the type of fruit but also in determining the ripeness of the fruit.
- Construct a pipeline from text to speech, which outputs the identification, ripeness and nutrition result as natural speech.
- Write well commented code, test demo, and a complete supporting document.

3.3 Rationale

CNN-based fruit and produce recognition is a well-known field of research, as public datasets are available, and deep learning approaches that classify fruits based on image data have been thoroughly demonstrated (Mureşan & Oltean, 2018). But there are few tools available that move beyond species identification. The objectives of this project are designed to fill that gap, and will integrate identification, ripeness grading, nutrition and usage information, and an entirely spoken output, all in one accessible pipeline, primarily for blind and low vision users.

4. Project Scope and Key Features

4.1 Scope

- Apple vs. banana image classification using a Convolutional Neural Network (CNN).
- System only detects 1 fruit per photo (not multiple fruits).
- There is no dedicated mobile app, and it is available via a regular mobile phone or laptop browser.
- No live video stream processing (only single photo capture).

4.2 Key Features

- CNN-based image classification that can be expanded to other types of fruits.
- Not a "low-confidence guess" but a confidence threshold that rejects images that are unclear or not recognised.
- Spoken (text-to-speech) announcement of the classification result.
- Ripeness detection in three stages: unripe, ripe and overripe.
- Information on nutrition, and usage returned per identified fruit.

5. Technical Approach & Resources

5.1 How the CNN Approach Works

The classification pipeline is written in the standard CNN pattern, where an uploaded image is pre-processed, then passed through convolution and pooling layers, followed by fully connected layers, and finally a SoftMax layer which outputs a class probability for apple and banana.

- Convolution: Filters move across the picture to detect features such as edges, shape, colour and texture to create feature maps (O'Shea & Nash, 2015).
- Pooling: Feature maps are down sampled to preserve only the strongest signals and to cut down on computation.
- Fully connected layers: the learned features are all combined into one representation of the image.
- Output layer: A SoftMax function outputs a probability associated with each class (apple, banana) – the greater the probability, the higher the likelihood that the prediction is correct.

5.2 Technology Stack

Table 1: List of technology stack

Category	Resources
Language	Python for the model and backend; JavaScript for the browser-based frontend.
Libraries	TensorFlow / Keras, OpenCV, FastAPI, pyttsx3 / Web Speech API for voice output.
Dataset	2,000+ self-collected photos, supplemented with Kaggle fresh/rotten fruit datasets.
Platform	Google Colab for model training; browser-based frontend; cloud or local backend server.

4. Project Management

4.1 Team Members

Table 2: Team roles, skill sets and task allocation

Team Member	Primary Responsibility	Supporting Responsibility
Ekta Chaudhary	CNN model development, training and validation; model evaluation and optimisation	Dataset preparation and collection
Riju Baral	Project coordination, implementation research, data preprocessing, testing and QA	Dataset collection
Samita Gautam	Documentation, technical support and integration; dataset organisation	Dataset collection
Sumeet Ghimire	Research on innovation, additional features and feasibility	Technical support and dataset collection

4.2 Timetable

Table 3: Twelve-week project timetable

Project Phase	Approximate Weeks	Main Focus
Planning and Research	Weeks 1–4	Topic selection, background research, requirements and planning
Data Collection and Preparation	Weeks 3–6	Collect, organise, label and prepare apple and banana images
Model Development and Training	Weeks 6–8	Develop, train, validate and refine the CNN model
Web Development and Integration	Weeks 7–9	Develop the browser workflow and connect the model with the application
Testing and Refinement	Weeks 9–11	Test classification and the upload-to-result workflow; refine issues
Documentation and Finalisation	Weeks 10–12	Complete documentation, final testing and project preparation

4.3 Time and Task Management

Progress is tracked in relation to the Week 1-12 timetable. The team is not waiting till the end of the project to see how things are going, they review milestones as the project progresses through the phases. This will enable delays or dependencies to be identified early on. If a member is behind schedule, tasks may be reworked among members so as not to unnecessarily delay the overall project.

Blockers are raised at the next scheduled team meeting so that they can determine an analysesate response. Tasks are also broken down by individual member's primary responsibility, and activities like collecting of datasets are shared. This way, each one is responsible, but the team helps each other when the job demands more.

4.4 Team Action Plan

The first step in the team action plan is to establish the scope of the project and conduct research to identify appropriate strategies. The need then is to gather, inspect and tag adequate number of apple and banana photos. After preparing the dataset, the team will develop a baseline CNN model and test it. The model will be developed followed by integration with the application which is browser based in order to upload an image and get the result delivered to the user from the model.

The comprehensive upload-to-result process will be tested and improved following integration. There will be ongoing documentation done throughout the project and not just in the final week. Expansion of the system will be prioritised after the core apple/banana classifier, and the associated fruit information is controlled to keep the project achievable in the available timeframe.

4.6 Meetings

Table 4: Weekly team meeting plan

Week	Meeting Focus
Week 1	Confirm the project topic, discuss the preliminary idea for the project and assign preliminary project research tasks.
Week 2	Discuss research findings, confirm project scope, key needs and roles of each team member.
Week 3	Discuss plans for projects and start organizing apple and banana picture material.
Week 4	Discuss the collected images and the quality of the data sets, image labelling and data preparation.
Week 5	Review data progress and discuss data pre-processing and data augmentation needs.
Week 6	Check out the first set of data and plan out for CNN model development and training.
Week 7	Review progress in development and training of models and discuss early results, blockers or technical issues.
Week 8	Go through the model performance and integrate trained model into the web application.
Week 9	Critically evaluate web development and test image upload, classification and result workflow.

Week 10	Run tests and quality assurance, detect faults and agree changes need to be made prior to finalisation.
Week 11	Check the full prototype, documentation and any unfinished work for the project.
Week 12	Perform final project review, review documentation and prepare for final submission and presentation.

4.7 Risk Management

Risk management applies to the agreed baseline: one uploaded image, apple/banana CNN classification, ripeness plus nutrition/usage information, browser delivery and spoken output. ISO 31000:2018 recommends identifying, analyzing, treating and monitoring risk, while PMBOK 7 emphasizes managing uncertainty to protect project outcomes (International Organization for Standardization [ISO], 2018; Project Management Institute [PMI], 2021). Risks will be reviewed during weekly progress checks and major phase transitions, with priority given to data quality, model generalization, scope control, and reliable end-to-end delivery.

Table 5: Project risk register and treatment plan

Risk, rating and project effect	Treatment / mitigation	Monitoring evidence
Limited/unrepresentative data (High). Too few or poorly varied images may reduce real-world performance.	Complete the planned 2,000+ images, remove poor/duplicate files, and check coverage across fruit, ripeness, lighting, angle and background.	Dataset count; class/ripeness distribution; sample-quality check.
Incorrect labels (High). Wrong fruit or ripeness labels can teach the CNN incorrect relationships and distort evaluation.	Use clear labelling rules, review ambiguous images, and peer-check a sample before freezing the dataset.	Label-audit log; corrected/removed image count.
Overfitting / weak generalization (High). The model may perform well on training data but fail on unseen images.	Keep separate training, validation and final test sets; compare training and validation behavior and use proportionate regularization/augmentation where required (Goodfellow et al., 2016).	Training/validation curves; unseen test results.
Low accuracy / unreliable confidence (High). Incorrect predictions could trigger misleading ripeness, nutrition, or usage of output.	Review class-level errors and retain a confidence-rejection mechanism for unclear images; set the threshold from validation evidence rather than an arbitrary value (Guo et al., 2017).	Class-level results; rejected-image tests; confidence review.
Scope creep (Medium). Adding extra fruits, live video, or a native app could delay the assessed baseline.	Freeze the baseline at one fruit per image, apple/banana classification, browser delivery, ripeness, information output and text-to-speech. Treat extras for future work.	Change log; weekly scope review.
Integration/demo failure (High). The model may work in isolation while uploading, preprocessing, backend, result of mapping or speech fails when integrated.	Test the complete upload-to-result path incrementally and rehearse it in the intended demo environment. ML systems require testing of software and data interfaces, not only the model (Amershi et al., 2019; Sculley et al., 2015).	End-to-end tests; resolved defects; successful rehearsal.

4.8 Resource Required

- Dataset: Images of fruits from public dataset, and personal images of apples, bananas and additional fruit taken under different lighting conditions and backgrounds.
- Software: Python, TensorFlow/Keras, GPUs (local or Google Colab).
- Libraries: TensorFlow/Keras for CNN and OpenCV for image pre-processing.
- Text-to-speech feature: Web Speech API, TTS or Google Cloud TTS for the spoken output

4.9 Development Methodology

Waterfall methodology is being used for this project because the scope is fixed at an early stage (apples/bananas, single image, baseline features) and each phase relies on the previous phase's completion:

Requirements/scoping, collection and labeling of datasets, design and training of CNN, integration with browser/TTS, testing and documentation.

It fits a small team with a fixed 12-week time frame and low re-planning requirements.

The project is feasible due to the careful management of the baseline around a single uploaded image and apples and bananas being the first fruit classes. CNN will be used in the proposed web application to classify fruit and provide useful information like ripeness, nutritional content, etc., and tips on how to use the item with speech as an accessibility feature.

The project will concentrate on providing a clean and representative data set, making sure that the model is evaluated and tested, and developing a reliable end-to-end workflow without distracting and unnecessary features until the baseline is achieved. Planned mitigation activities and planned monitoring activities for the main risks identified, which are limited data, incorrect labelling, overfitting, low accuracy, scope creep and integration or demonstration failure will manage the risks.

The team will apply complementary skills of model development, implementation of research, documentation, testing, and innovations in developing and testing the system. Once the agreed upon acceptance criteria are met and the entire workflow is working and tested, the outcome will be a working prototype that is tested and meets the project requirements and can be used as a foundation for further development.

Extensions planned include exploring further fruits, including berries, and further accessibility enhancements with text to speech and other capabilities. The prototype could serve as a foundation for various additional fruits and features that can be added in the future.

5.1 Next Step

The next steps convert the agreed scope into measurable completion evidence rather than adding new features.

Table 6: Next-step delivery plan and evidence of completion

Step	Required action and rationale	Evidence of completion
1. Finalize dataset	Complete, clean and label the apple/banana dataset; confirm class and ripeness coverage and permit use of any third-party images.	Frozen dataset and label check.
2. Train baseline CNN	Create separate training, validation and test sets, train the first model, and record performance. CNNs are appropriate for hierarchical image-feature learning (LeCun et al., 2015).	Saved model and recorded results.
3. Evaluate and refine	Review class-level errors, validation behavior and confidence; make only evidence-based adjustments to preprocessing, augmentation or hyperparameters.	Baseline-versus-revised comparison.
4. Validate upload workflow	Test upload, preprocessing, inference, ripeness/information mapping, result of display and text-to-speech, including unclear or unsupported images.	End-to-end pass/fail test record.
5. Prepare final demo	Rehearse in the intended environment, verify dependencies, document setup and limitations, and check for keyboard-accessible controls and readable/spoken results.	Demo checklist and successful rehearsal.

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